

Lesson 4 · Practice sheet

NCEA Level 2 · AS 91262 · Anti-differentiation, rates and kinematics

Time: 30 minutes

Answer version · tutor's working, not an official mark scheme

Show all working, including $+c$ and how you find it. Every rate and motion answer needs units. [A] Achieved, [M] Merit, [E] Excellence.

1. $f'(x) = 12x^2 - 2x + 3$ and $f(1) = 10$. Find $f(x)$. [A]

$$f(x) = 4x^3 - x^2 + 3x + c$$

$$f(1) = 4 - 1 + 3 + c = 10, \text{ so } c = 4$$

$$f(x) = 4x^3 - x^2 + 3x + 4$$

2. A particle has velocity $v = 8 - 2t \text{ m s}^{-1}$ after t seconds, and starts 5 m from the origin. Find its displacement at $t = 3$, and when it is at rest. [A]

$$s = 8t - t^2 + c; s(0) = 5, \text{ so } s = 8t - t^2 + 5$$

$$s(3) = 24 - 9 + 5 = 20 \text{ m}$$

$$\text{At rest: } 8 - 2t = 0, \text{ so } t = 4 \text{ s}$$

Question 3 [A]

- (a) A curve passes through the point $(1, 2.5)$ and has the derivative function: $f'(x) = 6x^2 + 5x - 1$

Find the co-ordinates of the point on the curve $y = f(x)$ where $x = 2$.

$$f(x) = 2x^3 + 2.5x^2 - x + c$$

$$f(1) = 2 + 2.5 - 1 + c = 2.5, \text{ so } c = -1$$

$$f(x) = 2x^3 + 2.5x^2 - x - 1$$

$$f(2) = 16 + 10 - 2 - 1 = 23; \text{ the point is } (2, 23)$$

Question 4 [A]

(b) For the function f :

$$f'(x) = 4 - 6x + 2x^2$$

The graph of $f(x)$ passes through the point (3,4).

Find the equation of the function f .

$$f(x) = 4x - 3x^2 + \frac{2}{3}x^3 + c$$

$$f(3) = 12 - 27 + 18 + c = 4, \text{ so } c = 1$$

$$f(x) = \frac{2}{3}x^3 - 3x^2 + 4x + 1$$

Question 5 [A]

- (b) An electric scooter is travelling down a straight footpath. Its velocity t seconds after passing a sign is given by $v(t) = 0.3t^2 + 1 \text{ m s}^{-1}$.

How far is the scooter from the sign when $t = 3$ seconds?

$$s = 0.1t^3 + t + c$$

$$\text{At the sign } (t = 0) \ s = 0, \text{ so } c = 0$$

$$s(3) = 0.1(27) + 3 = 5.7$$

The scooter is 5.7 m from the sign.

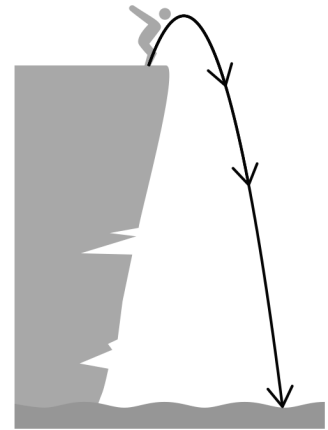
Question 6 [A]

- (c) (i) A cliff diver jumps up into the air above a cliff and then falls down into the water below.

Their acceleration is constant at -9.8 m s^{-2} .

The diver jumps up with an initial vertical velocity of 2.8 m s^{-1} .

Using calculus methods, find the velocity of the diver one second after they jumped.



$$v = -9.8t + c; v(0) = 2.8, \text{ so } v = -9.8t + 2.8$$

$$v(1) = -9.8 + 2.8 = -7 \text{ m s}^{-1}$$

(moving down at 7 m s^{-1})

Question 7 [M] · Excellence with full reasoning

- (ii) The diver hits the water at a velocity of -22.68 m s^{-1} (negative velocities indicate the diver is moving down).

Find the maximum height (above the water) that the diver reached.

You may wish to begin by finding the height of the cliff above the water.

Hits the water: $-9.8t + 2.8 = -22.68$, so $t = 2.6 \text{ s}$

Height above the water: $h = -4.9t^2 + 2.8t + H$, where H is the cliff height

$h(2.6) = 0$: $-4.9(2.6)^2 + 2.8(2.6) + H = 0$, so $H = 33.124 - 7.28 = 25.844 \text{ m}$

Highest point when $v = 0$: $t = \frac{2.8}{9.8} = 0.2857 \text{ s}$

$h = -4.9(0.2857)^2 + 2.8(0.2857) + 25.844 = 26.244$

Maximum height 26.2 m above the water

Question 8 (i) [A] · (ii) [M]

- (c) (i) During the cricket match, spectators both come and leave the venue. The number of spectators can be modelled by the following equation:

$$P = 100t^2 - 2t^4 + 750 \quad (0 \leq t \leq 7.5)$$

where P represents the number of spectators in attendance, and t represents the time in hours since the start of the match.

Show that the rate of change of spectators is -528 people per hour when $t = 6$.

$$\frac{dP}{dt} = 200t - 8t^3$$

$$\text{At } t = 6: 200(6) - 8(216) = 1200 - 1728 = -528$$

so the rate of change is -528 people per hour

- (ii) What is the meaning of this value for the rate of change?

At 6 hours after the start the number of spectators is decreasing

at 528 people per hour (people are leaving).